

Course Content for Final Exam

Contents/Topics	Question set	Marks distribution
Graphical representation of data: Construction of bar chart, histograms, Stem-leaf plots. Sample Space and Event: Tree diagram, Counting techniques	NW Exercise # 2.2 Questions 2.18-2.2,2.52,2.53,2.109 Exercise # 3.3 Questions 3.113-3.135 Questions 3.62-3.78 WP Questions 2.1-2.13 2.21-2.48,2.51,2.55,2.58 2.49-2.65 2.73-2.89 2.95-2.102 MB Questions 3.1-3.24, 3.28 3.1-3.24, 3.28 WP Questions 5.4-5.20, 5.25-5.28, 5.56-5.58,5.67,5.71 MB Questions 4.1-4.5,4.16-4.26 WP Examples 6.1-6.8 Questions 6.3-6.9	30%
Rules of Probability: Probability of an event, Additive rules, Multiplication Rule, Conditional Probability, Bayes' Rules		
Random Variables Concept of random variable and their types		
Discrete Random Variables and their distributions: Distribution of random variables: Main concept of random variables (CLO-1), PMF and CDF and Distribution of random vector: Joint and Marginal distributions and independence of random variables.		
Continuous Probability Distributions PDF and CDF, Joint Probability Distribution, marginal distribution, Conditional distribution (Bivariate)		
Types of random variables: Binomial, Poisson, Normal and standard normal distributions and applications		
Estimation: Introduction, confidence interval estimation using z & t distributions for single mean and difference between two means		
Hypothesis Testing: Testing of hypothesis for single mean and difference between two means using z-test and t-test Z(CLO-3), p-value method (CLO-3)		
Regression & Correlation: Scattered plot/diagram. Correlation coefficient, Introduction to simple linear regression, Relationship between correlation and regression, determining regression equation (the least square method) and Gradient Descent method w.r.t regression (material Will provide) (CLO-3), Inferences in regression coefficient , Inferences in correlation coefficient, Coefficient of determination.	WP Questions 9.1-9.5, 9.9- 9.13 and all solved examples related to the topic WP Questions 10.19-10.46 WP Questions 11.1-11.9, 11.11-11.14,11.5(a, b), 11.43,11.44,11.46 Examples 11.2-11.5 Questions 11.17-11.22 Questions 13.1-13.7 NW Questions 16.42-16.47	70%
Analysis of variance: Introduction to ANOVA , F-test, one-way ANOVA		